SIMATS ENGINEERING

ASSESSMENT TOOL 2 : Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 4

Code of Conduct

I, **C. Divya Charan (192525113)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Divya Charan

Assessment Tasks

Concept Mapping Tasks

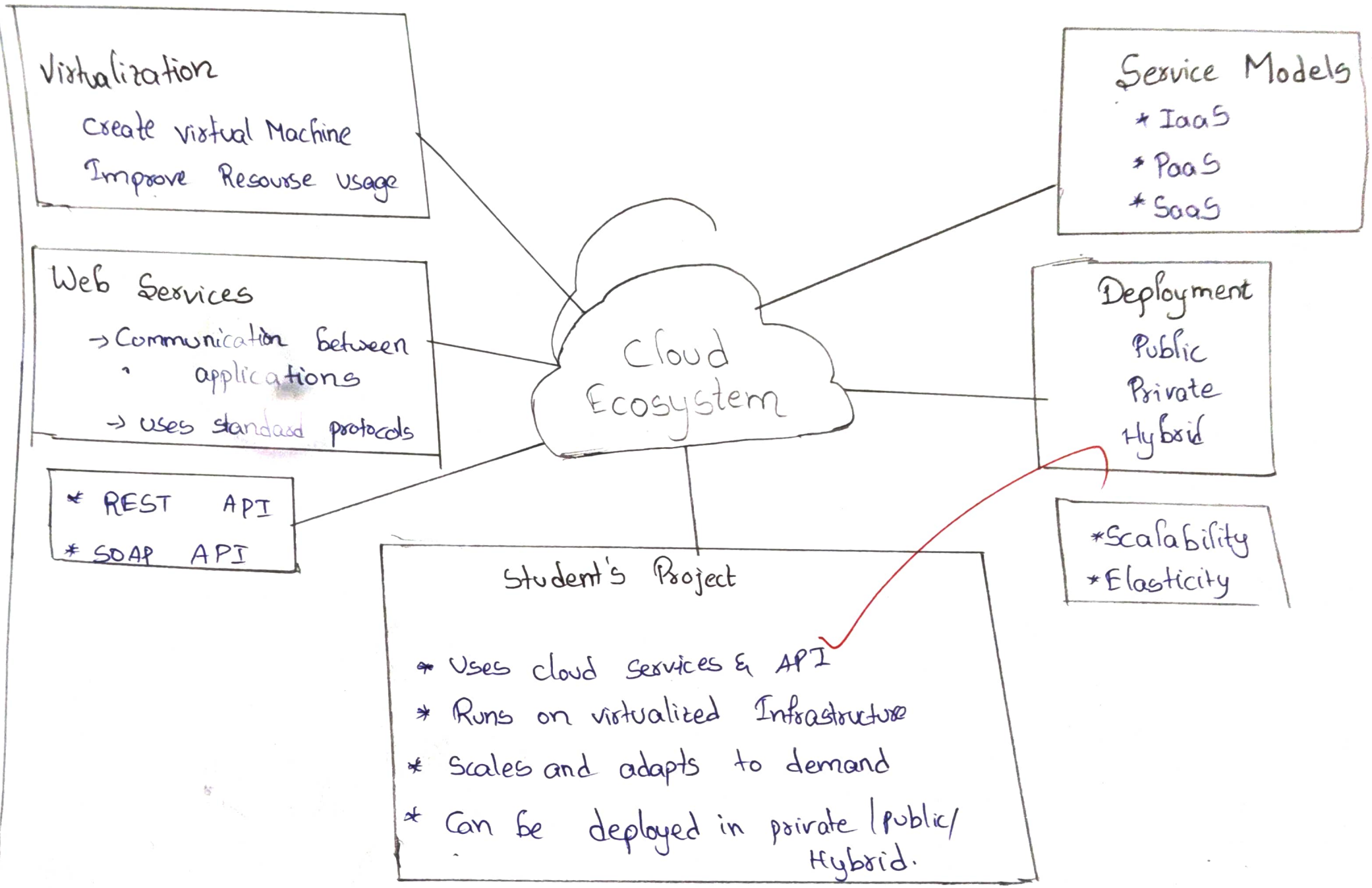
Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

1)

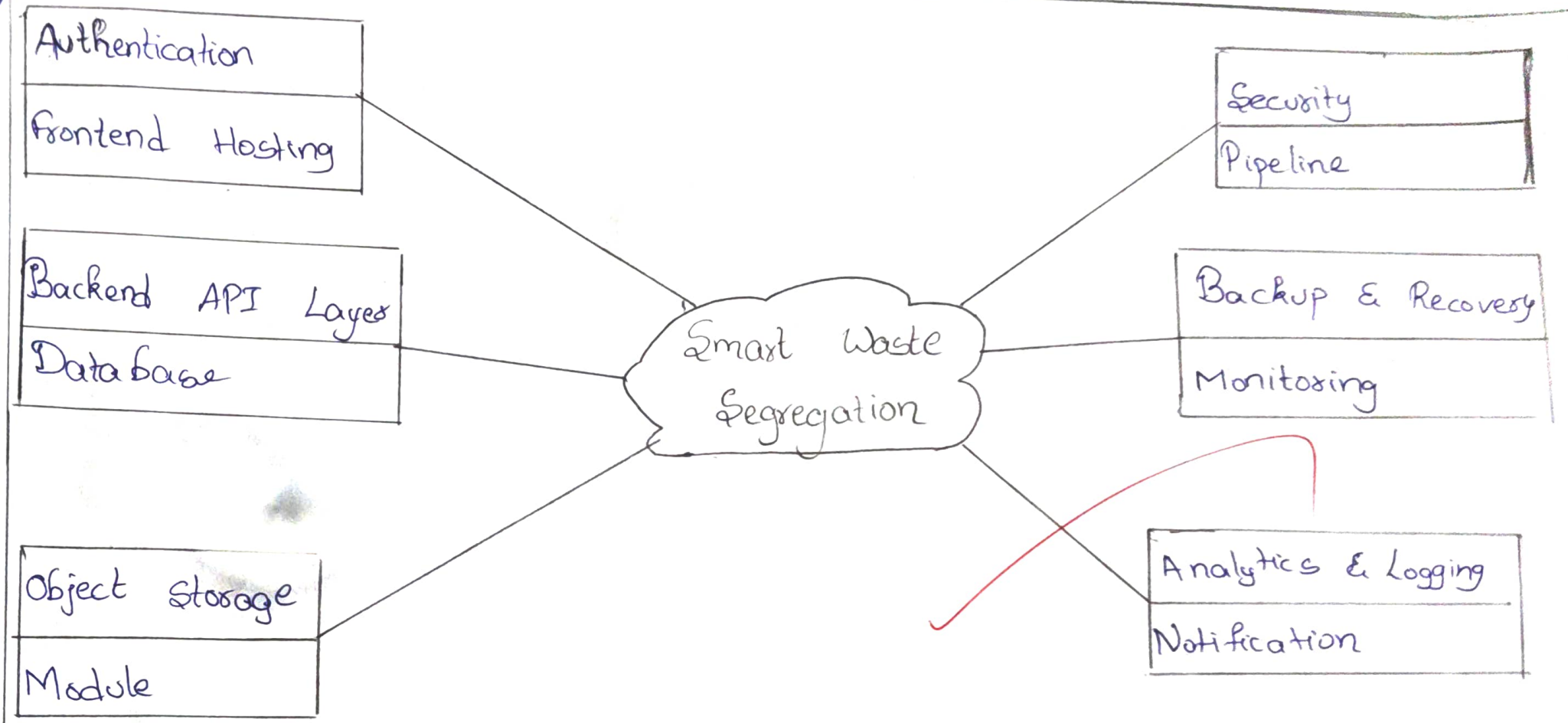

Virtualization
Create virtual Machine
Improve Resourse usage
Service Models
* IaaS
* PaaS
* SaaS
Web Services
→ Communication between applications
→ uses standard protocols
Cloud Ecosystem
Deployment
Public
Private
Hybrid
* REST API
* SOAP API
*Scalability
*Elasticity
Student's Project
* Uses cloud services & API
* Runs on virtualized Infrastructure
* Scales and adapts to demand
* Can be deployed in private / public / Hybrid.

1)

CLOUD SERVICE MODELS

Service Model	Description	Provider Responsibility	User Responsibility
IaaS	Provides Virtualized computing Resources	* Physical Data Center * Virtualization Infrastructure	* Create/manage virtual machines * Manage application
PaaS	Provides a managed environment to develop, run & deploy application	* Infrastructure * OS & runtime * Automatic Scaling	* Develop & manage code * Manage user data
SaaS	Provides Ready to use Software applications	* Complete application Software * Availability & Performance	* Using application * Manage accounts
XaaS	General term for various cloud Services.	* Underlying infrastructure Platform * Scaling, Patching & Availability	* Use & configure Service * Integrate with Application

3)


Authentication
Frontend Hosting
Backend API Layer
Database
Object Storage
Module
Smart Waste Segregation
Security
Pipeline
Backup & Recovery
Monitoring
Analytics & Logging
Notification

4)

Service Category	Managed by Provider	Managed by Student Team	WHY Suitable for Project
SaaS	Application Software	Configure and use	Ready to use Tools for Collaboration
PaaS	Server, OS, Runtime	Develop & Deploy	Deploy app without managing infrastructure.
IaaS	Pysical Server	VMs & Software	Custom Hosting
DBaaS	Database Infrastructure	Schema & Queries	Stores Records
STaaS	Storage Infrastructure	Files and access	Stores Reports
IDAaS	Authentication Service	Roles & Permissions	Secure Login
Monitoring	Monitoring Tools	Logs & Alerts	Track Errors

Student's Project Smart waste Segregation	* use cloud services * Stores Data & files in cloud * Support secure Access

13/10/26